# Cross-Generator AI-Generated Image Detection with CNN and CLIP Features

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***Abstract*—AI-generated image detectors often achieve high accuracy on images from generators seen during training, but may fail when evaluated on unseen generators. This project studies this cross-generator distribution shift on MS COCOAI, a 2026 dataset containing real images and synthetic images from modern generators including Stable Diffusion 3, SDXL, DALL-E 3, and MidJourney v6. I compare a ResNet-18 detector trained from scratch with a frozen CLIP ViT-B/32 image encoder followed by a linear classifier. With single-generator ResNet training, unseen-generator performance is weak in both directions: SD3 to MidJourney gives unseen AUC 0.5318 and MidJourney to SD3 gives unseen AUC 0.5482. Adding SDXL to SD3 training improves ResNet's MidJourney unseen AUC to 0.6694. The CLIP-linear detector improves all three protocols, reaching unseen AUC 0.9018 and F1 0.7096 for SD3+SDXL to MidJourney. These results show that generator diversity and pretrained visual-semantic features both improve robustness under generator shift.**

***Index Terms*—AI-generated image detection, cross-generator generalization, ResNet, CLIP, image classification**

## I. Introduction

Modern text-to-image systems can generate realistic images with diverse styles and semantics. This progress creates a practical need for detectors that can distinguish real photographs from AI-generated images. A major challenge is *cross-generator generalization*: a detector trained on one generator may learn generator-specific artifacts and fail on images from a different generator.

This project evaluates that failure mode using MS COCOAI [1], a recent dataset built from MS COCO images [2] and synthetic images generated by Stable Diffusion 3, Stable Diffusion 2.1, SDXL, DALL-E 3, and MidJourney v6. The goal is not only to maximize seen-generator accuracy, but to measure whether a detector remains reliable under generator shift.

The project makes three empirical comparisons. First, I train a ResNet-18 detector on one generator and test it bidirectionally across SD3 and MidJourney v6. Second, I test whether adding another training generator (SDXL) improves held-out MidJourney performance. Third, I repeat the same protocols with a frozen CLIP image encoder [3] and a linear classifier. The main finding is that ResNet-18 learns useful seen-domain signals but generalizes poorly, while CLIP-linear features substantially improve unseen-generator AUC and F1.

## II. Related Work

Convolutional neural networks such as ResNet [4] remain strong image-classification baselines because they learn hierarchical visual features directly from the training data. However, when trained from scratch on a narrow generator distribution, they can overfit to low-level artifacts such as texture, compression, and generator-specific rendering patterns.

CLIP [3] is pretrained on large-scale image-text pairs with a contrastive objective, producing image embeddings that capture broad visual-semantic structure. Because these features are learned from diverse web-scale data rather than a single fake-image generator, they may provide a more robust representation for AI-generated image detection under distribution shift.

## III. Method

### A. Problem Formulation

Each image $\mathbf{x}$ has a binary label $y \in \{0, 1\}$, where $0$ denotes real and $1$ denotes AI-generated. Both detectors produce two scores, one for "real" and one for "fake". A softmax layer converts these scores into a fake probability:

$$p_{\text{fake}}(\mathbf{x}) = \frac{\exp(s_{\text{fake}})}{\exp(s_{\text{real}}) + \exp(s_{\text{fake}})}. \tag{1}$$

During training, the model is penalized when the probability assigned to the correct class is low. This is the standard cross-entropy objective:

$$\mathcal{L}(\theta) = -\frac{1}{N}\sum_{i=1}^{N} \log p_{y_i}(\mathbf{x}_i). \tag{2}$$

Evaluation is performed on validation, seen-generator test, and unseen-generator test splits. The most important metric is unseen-generator performance.

### B. ResNet-18 Baseline

The first detector is a ResNet-18 classifier trained from scratch. Its calculation pipeline is:

$$\mathbf{x} \to \text{ResNet18}_\theta(\mathbf{x}) \to [s_{\text{real}}, s_{\text{fake}}] \to p_{\text{fake}}. \tag{3}$$

Images are resized and cropped to $224\times224$. The original ImageNet classification layer is replaced with a two-class real/fake head. All ResNet parameters are updated during training, so the network must learn both the image representation and the final decision boundary from the curated real/fake data. This makes it a clean test of whether a CNN trained from zero can learn generator-invariant fake-image evidence.

### C. Frozen CLIP Feature Detector

The second detector uses a frozen CLIP ViT-B/32 image encoder. Its calculation pipeline is:

$$\mathbf{x} \to g_\phi(\mathbf{x}) \to \mathbf{z} \to \mathbf{W}\mathbf{z} + \mathbf{b} \to p_{\text{fake}}. \tag{4}$$

Here $g_\phi$ is the CLIP image encoder and is not updated. It outputs a 512-dimensional image embedding, which is normalized before classification:

$$\mathbf{z} = \frac{g_\phi(\mathbf{x})}{\|g_\phi(\mathbf{x})\|_2}. \tag{5}$$

Only the linear head parameters $(\mathbf{W}, \mathbf{b})$ are trained. This separates representation learning from the downstream real/fake classifier. The reason for using CLIP is that its image encoder was pretrained with large-scale image-text contrastive learning. Instead of learning only local texture artifacts from one generator, CLIP starts from a feature space that already captures object identity, scene layout, style, and visual-semantic regularities. This should make the final detector less dependent on artifacts from a single synthetic generator.

## IV. Experiments

### A. Dataset and Protocols

MS COCOAI contains 96,000 real and synthetic datapoints generated from MS COCO captions and images. The experiments follow a staged logic:

- **Step 1a: ResNet SD3 → MidJourney.** Train from scratch on SD3 fakes and test on unseen MidJourney v6 fakes.
- **Step 1b: ResNet MidJourney → SD3.** Reverse the direction to check whether the transfer failure is symmetric.
- **Step 2: ResNet SD3+SDXL → MidJourney.** Add a second training generator to test whether generator diversity improves generalization.
- **Step 3: CLIP-linear comparison.** Repeat the same three protocols with frozen CLIP features and a trainable linear head.

For the single-generator protocols, the curated split contains 14,000 training images, 3,000 validation images, 15,000 seen-test images, and 15,000 unseen-test images. For the two-generator protocol, it contains 28,000 training images, 6,000 validation images, 30,000 seen-test images, and 15,000 unseen-test images.

### B. Implementation Details

All experiments run on an RTX 5090 GPU. ResNet-18 uses batch size 128, learning rate $10^{-4}$, weight decay $10^{-3}$, dropout 0.2, label smoothing 0.05, and early stopping with patience 2. CLIP-linear uses OpenAI CLIP ViT-B/32, batch size 256 for feature extraction, learning rate $10^{-3}$ for the linear head, weight decay $10^{-4}$, and early stopping with patience 8. Metrics are accuracy, F1, and ROC AUC. F1 and AUC are reported with fake as the positive class.

TABLE I
UNSEEN-GENERATOR PERFORMANCE. STEP 1 AND STEP 2 USE RESNET; STEP 3 REPEATS THE SAME PROTOCOLS WITH FROZEN CLIP FEATURES.

| Protocol | ResNet AUC | CLIP AUC | ResNet F1 | CLIP F1 |
|---|---|---|---|---|
| SD3 → MidJourney | 0.5318 | **0.8719** | 0.1747 | **0.6633** |
| MidJourney → SD3 | 0.5482 | **0.9153** | 0.1673 | **0.6045** |
| SD3+SDXL → MidJourney | 0.6694 | **0.9018** | 0.3695 | **0.7096** |

## V. Results

### A. Step 1 and Step 2: ResNet-Only Results

The first question is whether a CNN trained from scratch can learn a general fake-image detector. Step 1 tests this with a bidirectional single-generator transfer experiment. In Step 1a, ResNet is trained on SD3 fakes and evaluated on unseen MidJourney fakes. In Step 1b, the direction is reversed. These two runs are important because a robust detector should transfer in both directions, not only from one convenient source domain.

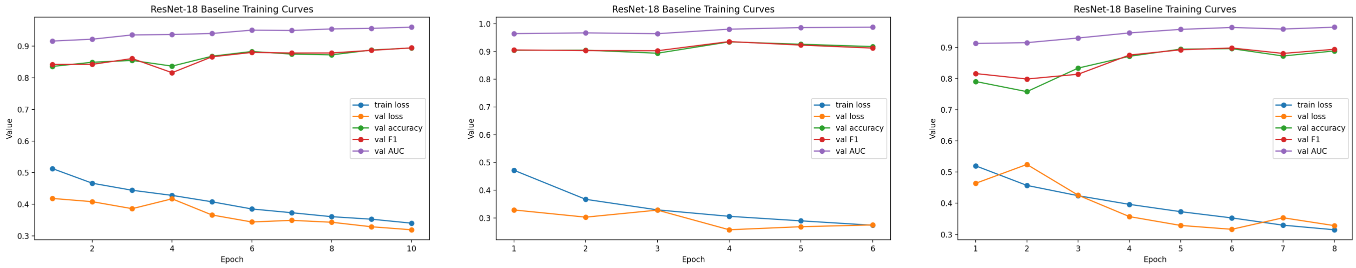
Figure 1 and Table I show that single-generator ResNet training does not generalize well. The unseen AUC is 0.5318 for SD3 → MidJourney and 0.5482 for MidJourney → SD3, close to random ranking. The corresponding F1 scores, 0.1747 and 0.1673, are even more concerning: the detector misses most fake images from the unseen generator. This indicates that the CNN has learned source-generator cues rather than generator-invariant evidence of synthesis.

Step 2 then adds SDXL to the SD3 training set. This raises ResNet unseen MidJourney AUC from 0.5318 to 0.6694 and F1 from 0.1747 to 0.3695. Therefore, generator diversity does help, but the improvement is limited. The detector is better than the single-generator version, yet it still fails to reach a level that would be reliable under practical generator shift.

### B. Step 3: CLIP Comparison

Step 3 is motivated by the limitation of the previous two stages. If a CNN trained from zero overfits to generator-specific artifacts, then the next question is whether a broader pretrained representation can provide more transferable features. CLIP is suitable for this role because its visual encoder is pretrained on large-scale image-text pairs. The model is encouraged to align images with textual concepts, so its image features encode higher-level semantic and structural information in addition to local appearance. In this project, CLIP is not fine-tuned; only a small linear classifier is trained on top of frozen CLIP image embeddings.

The difference is substantial. For SD3 → MidJourney, CLIP improves unseen AUC from 0.5318 to 0.8719 and F1 from 0.1747 to 0.6633. For the reverse direction, AUC increases from 0.5482 to 0.9153. In the strongest setting, SD3+SDXL → MidJourney, CLIP reaches unseen AUC 0.9018 and F1 0.7096. Figure 3 makes the improvement more direct: the ResNet split metrics still show a large gap between seen and unseen performance, while CLIP produces a more balanced pattern across splits.




(a) Step 1a: SD3 $\rightarrow$ MidJourney.

(b) Step 1b: MidJourney $\rightarrow$ SD3.

(c) Step 2: SD3+SDXL $\rightarrow$ MidJourney.

Fig. 1. ResNet-only training curves for the three staged protocols. The model can optimize the source task, but cross-generator performance remains weak, especially in the single-generator settings.

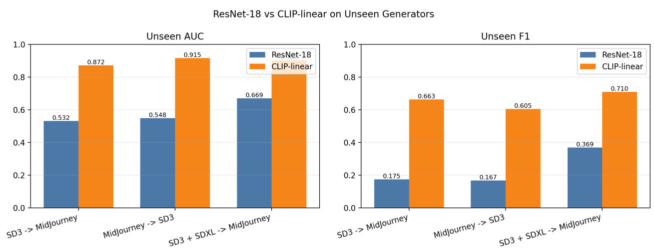



Fig. 2. Step 3 comparison on unseen generators. CLIP-linear improves AUC and F1 in all three protocols.

TABLE II
UNSEEN-TEST CONFUSION COUNTS. FAKE IS THE POSITIVE CLASS.

| Method | Protocol | TN | FP | FN | TP |
|---|---|---|---|---|---|
| ResNet | SD3 $\rightarrow$ MJ | 6984 | 516 | 6733 | 767 |
| ResNet | MJ $\rightarrow$ SD3 | 6323 | 1177 | 6708 | 792 |
| ResNet | SD3+SDXL $\rightarrow$ MJ | 6893 | 607 | 5663 | 1837 |
| CLIP | SD3 $\rightarrow$ MJ | 7249 | 251 | 3654 | 3846 |
| CLIP | MJ $\rightarrow$ SD3 | 7343 | 157 | 4183 | 3317 |
| CLIP | SD3+SDXL $\rightarrow$ MJ | 7315 | 185 | 3274 | 4226 |

Table II makes the error pattern more concrete. In the SD3-to-MidJourney protocol, ResNet detects only 767 of 7500 fake MidJourney images, while CLIP detects 3846. In the multi-generator protocol, ResNet improves to 1837 true positives, but CLIP still more than doubles that number to 4226 while keeping false positives low. The main gain is therefore not only better ranking AUC; CLIP also reduces the number of fake images that are mistakenly accepted as real.

### C. Qualitative Examples

Figure 5 shows fake MidJourney images from the unseen split where ResNet predicts "real" but CLIP predicts "fake". These examples make the numerical result more interpretable. ResNet appears sensitive to whether the target generator shares the same low-level artifacts as the training generator. CLIP, by contrast, can use broader cues such as object coherence, scene composition, and semantic plausibility. This does not mean CLIP is a perfect detector, but it explains why frozen CLIP features are more robust than a CNN trained only on the available synthetic sources.

### D. Limitations

This study evaluates a focused subset of MS COCOAI protocols rather than all generator combinations. CLIP is used as a frozen encoder only, so the results do not test whether end-to-end fine-tuning would further improve performance. The qualitative analysis is also limited to a small number of examples. Another limitation is that this project uses image-level binary labels only. It does not localize manipulated regions or explain decisions with saliency maps, so the current detector is best interpreted as a classification baseline rather than a forensic explanation system.

## VI. Conclusion

This project studied AI-generated image detection under cross-generator distribution shift. ResNet-18 performs reasonably on seen generators but fails to robustly detect unseen generators. Multi-generator training improves generalization, and frozen CLIP features produce the strongest results across all protocols. Future work should evaluate additional generator combinations, frequency-domain features, and CLIP fine-tuning.

## References

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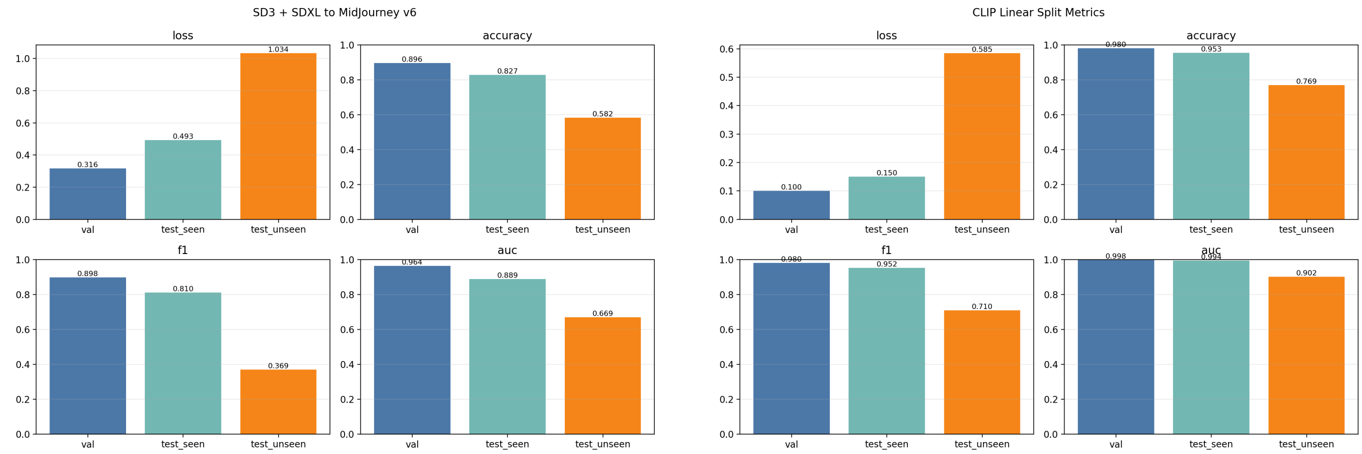
[4] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 770–778.




(a) Step 2 ResNet split metrics.

(b) Step 3 CLIP-linear split metrics.

Fig. 3. Direct comparison between the best ResNet setting and the corresponding CLIP-linear setting. CLIP keeps stronger unseen-generator performance while maintaining high seen-generator performance.

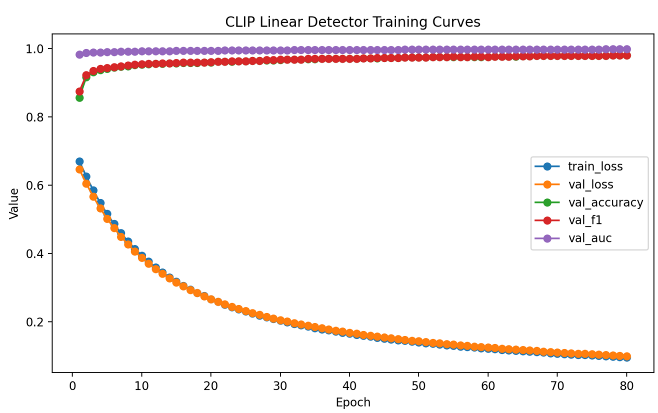



Fig. 4. Step 3 CLIP-linear training curves for SD3+SDXL $\rightarrow$ MidJourney. The linear head converges smoothly on frozen CLIP features.

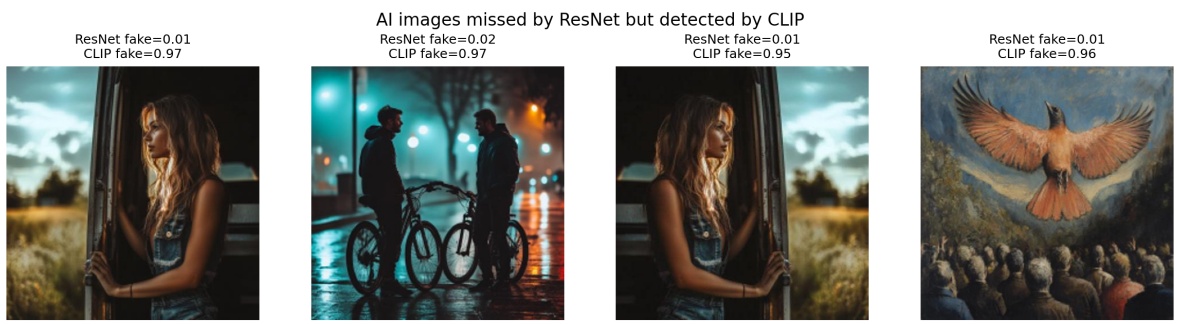



Fig. 5. Unseen MidJourney fake images missed by ResNet but detected by CLIP. Numbers are predicted fake probabilities.

## Appendix A
## Detailed Experimental Configuration

TABLE III
CURATED SPLIT SIZES USED IN EACH PROTOCOL.

| Protocol | Train | Val | Seen test | Unseen test |
|---|---|---|---|---|
| SD3 → MidJourney | 14000 | 3000 | 15000 | 15000 |
| MidJourney → SD3 | 14000 | 3000 | 15000 | 15000 |
| SD3+SDXL → MidJourney | 28000 | 6000 | 30000 | 15000 |

TABLE IV
MAIN HYPERPARAMETER SETTINGS.

| Setting | ResNet-18 | CLIP-linear |
|---|---|---|
| Backbone | ResNet-18 from scratch | frozen CLIP ViT-B/32 |
| Input size | 224×224 | CLIP processor default |
| Optimizer | AdamW | AdamW |
| Learning rate | $10^{-4}$ | $10^{-3}$ |
| Weight decay | $10^{-3}$ | $10^{-4}$ |
| Batch size | 128 | 256 feature extraction |
| Early stopping | val loss, patience 2 | val loss, patience 8 |
| Trainable head | two-class FC layer | linear real/fake head |

## Appendix B
## Implementation Notes

The ResNet baseline uses a two-class cross-entropy objective. The final fully connected layer is replaced with a real/fake head. The CLIP detector extracts normalized image embeddings once and caches them as NumPy arrays. A linear classifier is then trained on top of the frozen embeddings. This design keeps the CLIP experiment lightweight and makes the comparison focus on feature quality rather than full-model fine-tuning.

## Appendix C
## Code Archive and Reproducibility Commands

The complete project code is archived at:

https://github.com/Obl1vion31/aigc-detect

The appendix therefore does not duplicate the full source code. Instead, it records the role of each script and the commands needed to reproduce the main experiments. The experiments are organized under `outputs/current`. The main scripts are:

- `curate_mscocoai_split.py`: reads MS COCOAI parquet files, filters selected generators, and writes curated train/validation/test splits.
- `train_resnet18_baseline.py`: trains the ResNet-18 baseline from scratch and evaluates seen and unseen generator splits.
- `train_clip_linear_detector.py`: extracts frozen CLIP image features, caches them, and trains a linear real/fake classifier.

The following command pattern illustrates the three experiment types. In the actual runs, `TRAIN_GENS` is replaced by `stable_diffusion_3`, `midjourney_v6`, or `stable_diffusion_3,sdxl`; `UNSEEN` is the held-out generator.

```
python curate_mscocoai_split.py
  --train-generators TRAIN_GENS
  --unseen-generators UNSEEN

python train_resnet18_baseline.py
  --data-dir CURATED_SPLIT

python train_clip_linear_detector.py
  --data-dir CURATED_SPLIT
```

## Appendix D
## Qualitative Example Selection

The qualitative examples in the main paper are selected from fake MidJourney images in the unseen split of the SD3 to MidJourney protocol. They satisfy:

$$P_{\mathrm{ResNet}}(\text{fake}) < 0.5, \quad P_{\mathrm{CLIP}}(\text{fake}) \geq 0.5. \tag{6}$$

Thus, each example is a ResNet false negative but a CLIP true positive. This selection rule is used to show the specific error type reduced by CLIP: fake images that a source-trained CNN would incorrectly accept as real.

## Appendix E
## Core Pseudocode

This section summarizes the key computation without reproducing the full code. The CLIP detector freezes the encoder and trains only a linear head:

```
z = clip_image_encoder(x)
z = normalize(z)
logits = linear_head(z)
loss = cross_entropy(logits, y)
```

The ResNet baseline trains the classifier end-to-end:

```
logits = resnet18(x)
loss = cross_entropy(logits, y)
loss.backward()
optimizer.step()
```